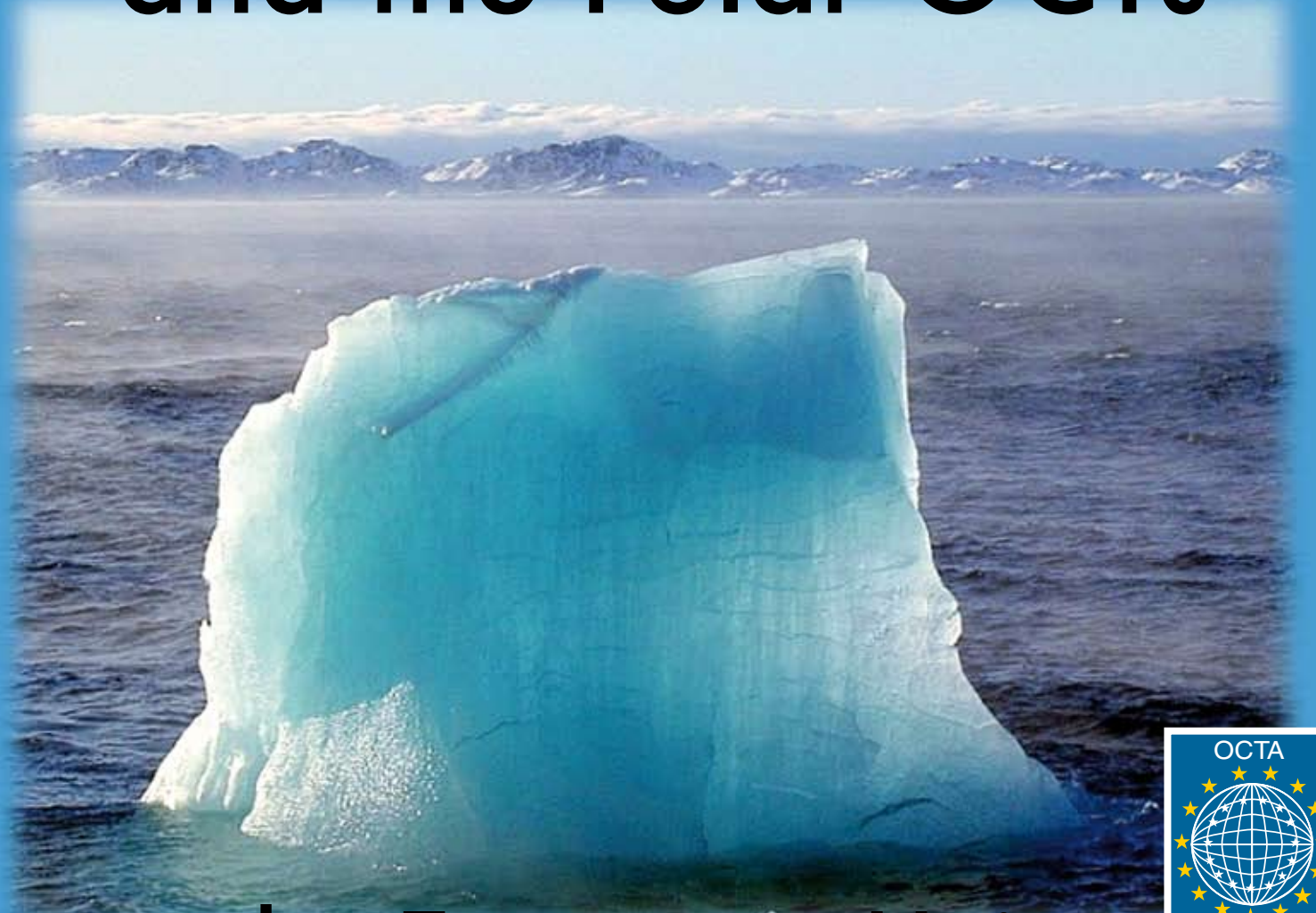


From the Tropics

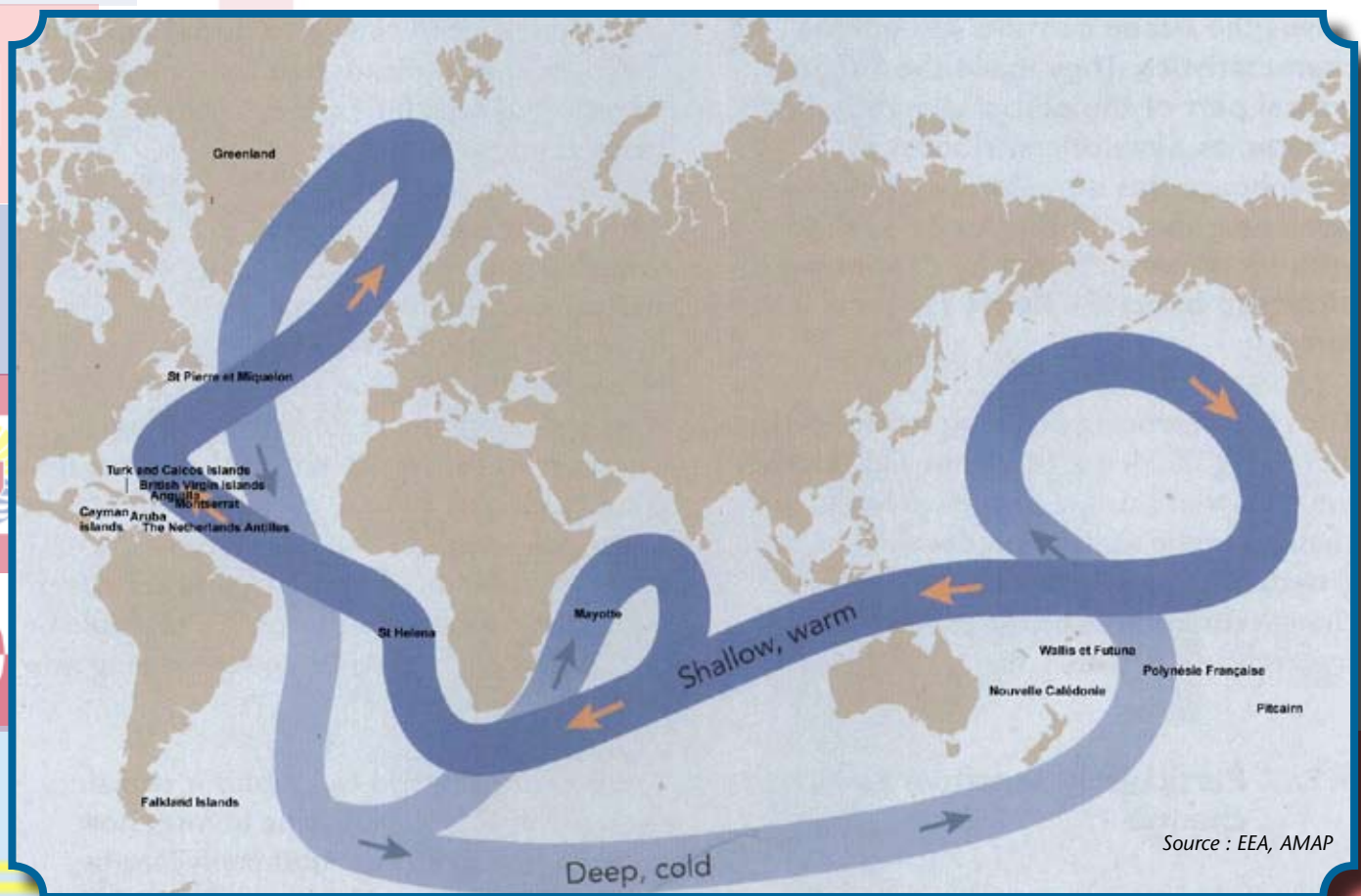


and the Polar OCTs



to the European Union

OCTs and Global ocean circulation



Association of Overseas Countries and Territories of the European Union (OCTA)

The Association of Overseas Countries and Territories of the European Union was established as a non-profit association in Brussels, Belgium, on March 5, 2003, pursuant to an Inter-governmental Agreement signed between 13 Overseas Countries and Territories on September 18, 2002.

The Articles of the Association of OCTA detail the Objectives of the Association and these include defending the collective interests of the members and representing these interests vis-à-vis the institutions of the European Union for all matters outlined in the OCT Decision (www.octassociation.org).

This brochure will highlight an issue of common and collective interest : our oceans and the environmental impact that climate change has on these oceans. Social and economic development and the protection of the environment are inseparable pillars of sustainable development. Moreover, as small islands developing countries and territories, the OCTs are severely constrained by adverse factors such as small size and limited resource endowment. This is further compounded by the adverse consequences of climate change on their fragile ecosystems. The OCTs are therefore calling for a sustained global initiative to combat climate change and for urgent attention to be given to the special needs of the OCTs in coping with the impact.

Nine OCTs contributed to this brochure. However, the issues highlighted are common for OCTs as a whole.

INTRODUCTION

Climate change from the poles to the tropics

OCTA is the association of overseas countries and territories (OCTs) of Denmark, France, The Netherlands and United Kingdom.

The OCTs are scattered all over the world and vary greatly in size and climate, from great cold Greenland in the north to the tropical archipelagos of French Polynesia in the South Pacific.

Oceans and seas play a key role for the OCTs as they are a primary source of the OCTs' economy. Climate change and its effect on the oceans and the environment will significantly affect the OCTs.

Human activity speeds up the process of climate change. The temperature of the ocean is rising, causing the melting of polar ice, the dilation of the seas, and an increase in the risk of tropical storms. The rise in sea temperature is also endangering the coral reefs, which will be less effective at protecting coastlines from increased wave activity.

OCTs find combating and managing climate change issues particularly difficult because they only have limited resources. Oceans and seas must be managed in multilateral fora and for OCTs in cooperation with the EU.

Since the ocean is a common reference for the OCTs, this pamphlet illustrates the impact of two concrete oceanographic consequences of climate change, namely the heating of the oceans and the rising of the sea level. Climate change is a fact and a strengthened effort is urgently needed.



CLIMATE CHANGE IN ARUBA



Aruba is located in the heart of the southern Caribbean, 15 miles (20 kilometres) off the coast of Venezuela. It is 19.6 miles (30 kilometres) long and 6 miles (9 kilometres) across, at its widest point, with an area of approximately 70 square miles (184 square kilometres).

The average temperature is 82°F (28°C), with cooling trade winds and the rarity of tropical storms and hurricanes. Aruba lies at the boarder of the hurricane belt and therefore the rainfall is very moderate, averaging about 16.1" (408.9 mm) per year.

The rainfall season of the year 2005 again was wet to very wet. Aruba received 91% above normal rainfall from October through December. This is the result of above normal sea surface temperatures in The Caribbean Sea and Tropical Atlantic and neutral sea surface temperature conditions in the Pacific Ocean.

Aruba has a population of about 100,000 inhabitants, with some 700,000 visitors a year.

THE RISING OF THE SEA LEVEL

The economy of Aruba is heavily dependent on tourism. The closing of refinery in 1985 was an economic setback for the Aruban community. Faced with rising unemployment, the island directed its full energies into the tourism industry. The successful development of the island's tourism product resulted in a three-fold jump in the number of visitors, increasing from 206.747 in 1985 to 641.906 in 2003. Today, tourism is the main pillar of the Aruban economy. Climate change, particularly concerning sea level rise in the coastal zones, is therefore hugely relevant to Aruba.

Our most vulnerable areas to sea level rise are the beaches and salt margins on the southwest side of the island. The salt margins are our natural flooding areas but these are under pressure to be extended in order to develop the tourist industry.

Due to urbanisation man-made civil works obstruct some of the riverbeds. More hard surface (roofs, roads and pavement) in combination with increase in rain results in heavier rainwater run off.

INITIATIVES ALREADY TAKEN BY THE ARUBAN GOVERNMENT

Aruba has a multi annual Sustainable Economic Development Policy Plan, some of whose projects and programmes address climate change. Climate change is not a high priority on the local agenda though. However a positive development is that multi-stakeholders meetings on different subjects about sustainable development are held more frequently.

The Aruban Government is considering the ratification of the United Nations Framework Convention on Climate Change and the Kyoto Protocol.

As a small island partially dependent on the refinery and totally dependent on the Water and Energy Plant the ratification of the Kyoto Protocol is particularly important to Aruba. Next to the two main point sources of air pollutions we have traffic as the biggest non-point source. And we are challenged by the demand in more mobility, meaning more asphalt and cars, in the short term.

To monitor the air quality Aruba has a multi annual plan in place and envisions having this monitoring programme running by the first quarter of 2007.

After having received a tail of Hurricane Lenny in 1999 and the passing by of Hurricane Ivan in 2004 Aruba's response was to improve the rainwater run off system. A multi annual plan (2006-2009) has been drawn up to address this.

Furthermore the government owned Water and Energy Plant is planning to extend its electricity production with windmills.

The Aruban government has established the Coastal Zone Management Institute to manage and protect the Aruban coastal areas. A calamity plan has also been developed.

THE NEED FOR FURTHER ACTION

Current knowledge and available techniques are not able to provide robust quantitative indications of potential changes in hurricane frequency, intensity or if the hurricane belt will shift. Aruba is eager (as are other Caribbean islands) to have conclusive scientific research on the implications of climate change in the Caribbean.

With regard to climate change, Aruba is a small island with big challenges.



CLIMATE CHANGE IN THE CAYMAN ISLANDS

The Cayman Islands are located 470 miles southwest of Miami, 150 miles south of Cuba and 170 miles northwest of Jamaica. They are comprised of Grand Cayman, Cayman Brac and Little Cayman with a total population of approximately 45,000. Grand Cayman is the largest and most populated of the three and the capital, Georgetown, is home to the main business centre.

The Islands are tops of pinnacles reaching up from the Cayman Trench – one of the deepest oceans in the world. The three islands are formed of two distinct formations of calcareous rock. The older limestone, called bluff limestone, was formed in the Oligocene-Miocene period, about 30 million years ago. This limestone forms the central core of each island. It is a dense Karst limestone. Surrounding this bluff limestone is a coastal limestone terrace called “ironshore”. The ironshore is a formation of carbonates created by living organisms such as corals, algae and shells which were laid down during sea level changes over the past 30 million years.

THE RISING OF THE SEA LEVEL

Present extreme high water levels under storm surges, tides and waves would be attained more frequently if the mean sea level rises.

Sea-level rise will cause salt-water intrusion of freshwater lenses thereby reducing their ability to support biodiversity and agricultural needs.

Sea-level rise over the next century is likely to outpace the vertical growth rate of coral, which

will be deeper, receive less sunlight and grow more slowly. In turn, coral reefs will be less effective at protecting coastlines from increased wave energy and will produce smaller amounts of reef sediments that build and support beaches. Sea-level rise will also challenge the ability of mangroves to keep pace through inland migration. In addition, optimal water depths for turtle grass and other important sea grass species may be exceeded by sea-level rise. These systems are already under stress from coastal development and recent hurricane impact.

Select beaches currently support waning nesting populations of endangered green and loggerhead turtles. Many of these beaches are low and narrow making them more susceptible to sea level rise. The vulnerability of nesting populations will increase with rising seas as beach area for nesting will be reduced. Many beaches do not have the potential to migrate inland because of encroaching coastal development due to poor building practices, making them unsuitable in future to support nesting activities.

Sea-level rise is receding coralline beaches worldwide and exacerbating beach erosion events. Public access along the beach where seawalls and other artificial obstructions have been placed will become an even greater issue. Tourism revenue will be lost by beachfront facilities and the country as a whole if beach amenities are compromised by user conflicts on ever narrowing beaches. Ability of the beach to flex will be that much more reduced with rising seas and stationary coastal construction.

THE HEATING OF THE OCEANS

Sea surface warming rates over coral reef regions such as the Caribbean have been shown to be higher than the global average.

The minimum increase in global sea surface temperature of +1°C is expected by the end of 2100 with more alarming projections of 1°C by 2015 and about 2°C for the 2050s for the Caribbean Sea.

A rise in sea temperature of 1°C will have significant impact on reef systems locally, as manifested during a coral reef bleaching event throughout the Caribbean and Atlantic in 1998 that lasted from August to October.

Bleached corals are susceptible to other coral diseases and diseased coral is often out-competed on spatially-limited reefs by algae.

The future reef ecology may result in a shift in coral community and cover with associated impacts on the structure of fish communities and other marine species. Therefore bleaching events and their impacts can have devastating implications for dive tourism, which accounts for a significant portion of tourism revenue.

Ocean temperatures provide fuel for tropical storms under optimal conditions. Warmer sea surface temperatures may gradually increase the risk of highly destructive category 5 storms

occurring as increases of up to 10 to 20% in tropical storm peak wind and peak rainfall intensities are projected. As maximum wind speeds of the strongest hurricanes rise so too will insured losses throughout the islands as successive storms will further reduce the ability of natural buffers like coral reefs, fringing mangrove and tree lines to perform protective functions. Some areas which are relatively flood-free may become flood-prone due to increased precipitation associated with more intense storm events.

Physical and biological impacts on coral reefs from rising sea temperatures and tropical storms can have deleterious effects on certain fish communities. Emerging evidence suggests that fish communities are also negatively affected by the post-impact response of mangrove systems.

THE NEED FOR FURTHER ACTION

Government insurance coverage and expenditure will have to plan for these increased risks, which may well reduce the earning potential of the country. Natural disaster management, including securing adequate funding for recovery efforts, will necessitate the incorporation of both disaster mitigation planning and adaptation strategies to lessen these risks.



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CLIMATE CHANGE IN THE FALKLAND ISLANDS

The Falkland Islands are an archipelago made up of over 700 islands with a total land mass of 12,200 km sq (1.2 million hectares) and are located in the South Atlantic Ocean, roughly 500 kilometres east of the southern tip of South America.

With a temperate climate, the Falkland Islands enjoy a narrow temperature range between a minimum of around -5° in July to a maximum of 24° in January. Rainfall is relatively low with an average of 24 inches (626mm) per year, while strong winds are common (averaging 15 knots).

With a civilian population of approximately 2,500 and a military population of 1,200, the Falkland Islands are an Overseas Territory of the United Kingdom. Islanders enjoy a high standard of living due to the development of a successful fishing industry since 1986. Due to the low population and absence of any heavy industry, carbon emissions produced in the Falkland Islands are minute in comparison to other countries, but the largest source of emissions is likely to be the 580,000 sheep farmed on the Islands.

The economy of the Falkland Islands is heavily dependent on the rich fishery grounds around the Islands. A successful fisheries industry, primarily based on squid, has been operated and managed by the Falkland Islands Government (FIG) for the last twenty years. Climate change, particularly concerning oceanography, is therefore hugely relevant to the Falkland Islands. Any dramatic ocean temperature changes could have adverse implications for the Islands' fisheries and economy as a whole.

FIG policies and programmes addresses climate change. The most significant and recent document is the Draft Conservation and Biodiversity Strategy (CBS). This identifies current trends for the biodiversity of terrestrial and vast marine communities of the Islands. The CBS also focuses on how to target limited resources to address the risks to its biodiversity.

A workshop was convened with stakeholders, to establish risks to biodiversity. Actions were

then prioritised according to the level of success which they might result in. Climate change was ranked as a high risk to the biodiversity of the Islands. The following text is an extract from the Draft CBS.

"Climate change is widely considered to be one of the most serious adverse impacts on native biodiversity. This is especially true for islands such as the Falkland Islands because species and associated habitats have little 'room for manoeuvre' in terms of latitudinal shifts. Even in areas where latitudinal shifts are possible, it is possible that rates of climatic change may exceed the ability of species to adapt and move. The likely impacts of climate change in the Falkland Islands are largely unknown and though the standard global circulation models provide a coarse scale estimate of temperature and precipitation changes, the impact on ocean circulation and how this will affect Falkland Island marine and terrestrial communities is unknown.

" Mitigation of climate effects is wholly beyond the ability of Falkland Islands to implement, and the only means is through international protocols such as Kyoto. However, potential action for FIG includes a better understanding of climate impacts on biodiversity as well as the principal industries that dominate the economy, notably agriculture and fisheries as well as the social aspects of life in the Falkland Islands."

INITIATIVES ALREADY TAKEN BY THE FALKLAND ISLANDS GOVERNMENT

FIG agreed to join the United Kingdom's ratification of the United Nations Framework Convention on Climate Change and the Kyoto Protocol, in April 2006. As the second British Overseas Territory to join the UK's commitment to the UNFCCC and Kyoto, this represents a willingness to promote environmental concerns.

The main environmental development with the FIG joining the Kyoto Protocol will be the monitoring of emissions, on an annual basis. The agreement with the UK stipulates that the FIG is



expected to introduce, where possible, policies in line with the objectives of the UK Climate Change Programme. Although emissions reductions are not specifically required, there is goodwill, and FIG is already pursuing several emissions-reducing projects.

A Waste Heat Recovery programme is being implemented by the Government Power Station, which will use otherwise wasted electricity to heat the local hospital and community school. The scheme will come online in early 2007, and will displace annually the use of about 300,000 litres of diesel (displacing roughly 3% of annual fossil fuel consumption for the power station).

FIG is considering the installation of a wind farm, which would be the first large scale renewable energy supply for the Falkland Islands, and would go further to displace fossil fuel use. Predictions suggest a displacement of over 10% of annual power station diesel consumption.

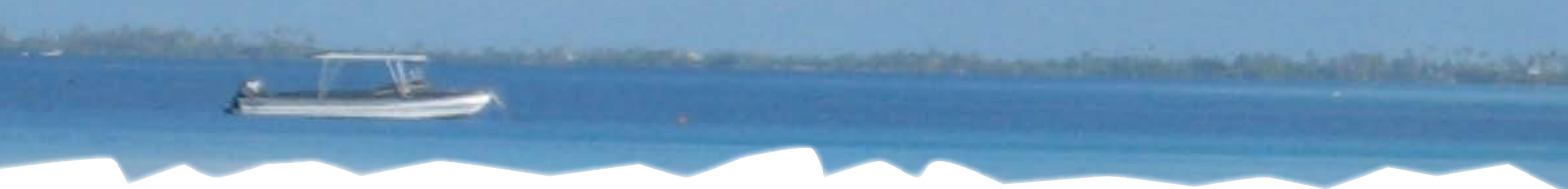
On a smaller scale, the provision of wind turbines for small settlements in the Falkland Islands has provided wind-generated power to small homesteads for the last ten years. The Falkland Islands Development Corporation (Government-funded development agency) offers a 50% grant on wind turbine costs to farmers.

Building control guidelines are being introduced by the FIG Environmental Planning Department to promote the building of more energy efficient housing.

CONCLUSION

With regard to climate change, the Falkland Islands Government feels it is 'acting locally, thinking globally'.





CLIMATE CHANGE IN FRENCH POLYNESIA

French Polynesia lies in the southern Pacific Ocean, 18,000 km away from metropolitan France, 6,500 km away from the United States and 5,900 km from Australia. It covers almost 5,500,000 km² within the Exclusive Economic Zone, which is an area equivalent to Europe. It consists of 118 high-lying volcanic islands and low-lying coral island atolls, whose highest point is no more than 1.5 metres. The islands are grouped around five archipelagos – Société, Iles du Vent, Iles-Sous-le-Vent, Tuamotu-Gambier, Australes and Marquises. Within this large territory, the ground above sea level only covers 3,430 km² and the lagoons about 12,800 km² - that is, 0.06 % of the earth's surface.

Polynesian reefs offer geomorphological diversity which is remarkable because of the great number of types that exist there. These reefs form an extremely rich, specifically coastal intertropical ecosystem, on a par with that of tropical rain forests. Polynesia's reefs offer a unique diversity of types – fringing, barrier, atoll and patch. In the Société archipelago, the development of a sparsely colonised volcano, Mehetia, into an atoll, demonstrates in a particularly striking way the theory of how atolls came into being. The diverse species of marine life include 1,500 species of molluscs, 350 species of algae, 170 species of coral and 800 of fish.

This wealth of species is in great danger. French Polynesia has more animal and plant species already extinct or threatened with extinction than any other overseas territory.

THE RISING OF THE SEA LEVEL AND THE HEATING OF THE OCEANS

The scientific world may well have been studying global warming and its effects for twenty years or so. However, it remains a fact that French Polynesia started taking this problem to heart only very recently. As at 2005, the National Bureau on Climate Impact had listed only five studies^{1, 2} on the subject. Moreover, none of

them mentioned any foreseeable impact on the region. The scanty information obviously made it impossible to determine the extent of the problem. However, some studies are currently underway, in particular a thesis on the consequences of climate change for French Polynesia's economy, so we should soon be have this eagerly awaited data³.

The archipelago of the Iles du Vent has the highest number of inhabitants. There are two islands, out of 118, that house three quarters of French Polynesia's inhabitants. The urban area of Papeete covers about 40 square kilometres, i.e. one percent of French Polynesian territory, accounting for more than 65% of the population¹. Indeed, this disparity may grow even more pronounced, as the sea level rises and the coral atolls and the most low-lying islands are submerged. Also, emigration into countries across the border may be on the cards, as has happened in some island states (cf. Organisation of Small Islands)

Climate changes linked to man's activities on these islands have been very rapid. Indeed, such changes considerably threaten the region's fragile ecosystems. Moreover, natural resources will be strongly and equally affected, with a knock-on impact on both the local and regional economy, which that rely on the island's wealth of animal and plant life as a source of income.

From 1964 onwards the traditional economy of Polynesia, which until then had been essentially agricultural, underwent a profound change when the Centre d'Expérimentation du Pacifique was set up. Ever since then the Gross National Product has increased considerably, bringing about profound changes to society and more particularly a migratory flow to Tahiti. Whereas



¹) See census pertaining to a study on the impact of climate change on coastal areas in the French DOM-TOM Overseas Territories, ONERC, French National Bureau on Climate Impact, 2005

²) See, assessment on the degree of climate change, its causes and foreseeable impact on French geography on the eve of 2025-2050, 2100, Senate, 2002

in 1964, 59% of the population in jobs worked in the primary sector, today only tourism and black pearl culture, vital economic activities and directly related to climate change, have helped to improve the terms of trade. After twenty years, tourism has improved by almost 110% - in 2004, about 212,000 tourists visited Polynesia⁴. Moreover, maritime and lagoon resources are an economic factor that should not be overlooked:

- By and large, the export figures for cultured pearls today account for ten billion of FCP's turnover, i.e. half of exports.
- Despite the informal, indeterminate nature of lagoon fishing, almost every Polynesian is involved in it socially, economically and culturally, making it as vital for Polynesia as deep sea fishing. Indeed, this kind of subsistence fishing is often an indispensable source of income for isolated populations.

Even though French Polynesia does not have detailed scientific data on the consequences of a rising ocean level for tuna fish stocks and other species that serve as a source of food, or on mother-of-pearl cultivation, it is inevitable that a climate change such as this will have a damaging impact on these resources, on the farming industry and, consequently, on the economy as a whole.

Moreover, the majority of companies in Tahiti are located on the coast or in the valleys around major urban centres. By their very nature, the rising waters will likewise entail rapid changes and necessitate a relocation and adaptation of French Polynesia's industrial activity, which is almost at sea level.

THE NEED FOR FURTHER ACTION

Faced with this situation French Polynesia, like many other states in the Pacific, has no choice but to turn towards and work in co-operation with

privileged partners such as the European Union.

French Polynesia unveiled its strategy to combat climate change at the Conference on Climate Change, first held in Montreal in December 2005.

However, this European Union's biodiversity hotspot wishes to reach out beyond activities at local level. French Polynesia wishes to associate itself with the steps initiated by the European Union's Council, Commission and Parliament within the framework of the Sixth European Programme on Climate Change and the implementation of the instrument LIFE +. This would highlight the priority given to the issue, raising awareness on a very large scale and influencing public opinion at local, European and international level.

We are determined to be politically strong willed and work in partnership with all the actors involved, bearing in mind how great the stakes involved in climate change are.



Agglomération de Papeete, Initialisation du modèle.



Résultat de la modélisation après stabilisation du niveau d'eau dans le cadre d'une élévation du niveau de la mer. Juin 2006 ³

³) Impacts du Réchauffement climatique sur les petites îles du Pacifique, modélisation et perception du risque PRODIG, unité de recherches CNRS au sein de l'Université de la Sorbonne et IRIDIP, unité de recherche au sein de l'université de la Polynésie française, juin 2006

⁴) See the statistic of Institut de Polynésie française



CLIMATE CHANGE IN GREENLAND

Greenland, which is a Danish OCT, is the biggest island in the world and is situated right next to the North Pole. Greenland covers 2.2 mill. km², 81 % of which is covered with ice.¹

“Ice sheet” is the designation of a collection of ice caps and glaciers, which are land-based ice covering respectively mountains and valleys and are contrary to “sea-ice”, which is formed when the seawater freezes.²

At the centre the Greenlandic ice sheet is up to 3 km thick. The thickness decreases towards the ocean and on the fringes it is only a few hundred metres thick.

The thickness of the ice sheet is slightly increasing, but the surface area as a whole is decreasing as the ice is melting or “calving” on the fringes, meaning that chunks of ice are breaking off the sheet. In total the ice sheet is losing its density.

Climate change in Greenland due to human activities around the world causes this process to take place at an accelerating speed, entailing problematic consequences such as the rising of the sea level and the heating of the oceans.

THE RISING OF THE SEA LEVEL

During the 20th century the global and the arctic sea level has risen 10 – 20 cm, and this evolution might accelerate if no action is taken towards the rising temperature of the oceans³. If the Greenlandic ice sheet melts totally the global sea level will rise 7 metres.

THE HEATING OF THE OCEANS

A warmer ocean environment will result in changes in the fish stocks.

The fisheries industry is the key industry in Greenland and fish products account for 86 % of all export. Therefore any changes herein will greatly influence the Greenlandic economy and thereby society as such.

Because of climate change the need for an alteration of the industry due to changes in the fish stocks is not unthinkable, indeed may be expected.

The sea ice is thinning. The annual average sea-ice extent has decreased by about 8% during the past 30 years⁴. This is a two-edged sword in the traditional hunting and fishing culture, as it shortens the period where it is possible to hunt and furthermore reduces the habitat for polar bears and ice-habiting seals driving these species towards extinction. The foundation of this culture is thus slowly disappearing.

INITIATIVES ALREADY TAKEN BY THE GREENLANDIC GOVERNMENT

Greenland is subject to the Kyoto protocol and is thus obliged to further its objectives of reducing CO₂ emissions.

In these days of increasing energy consumption, hydropower plants provide a sustainable and long term solution to the problem of reducing



¹) See Statistics Greenland

²) See Arctic Climate Impact Assessment (ACIA) a report from 2004 by the Arctic Council of Ministers, www.acia.uaf.edu

³) See ACIA

⁴) See ACIA



CO₂ emissions. Some Greenlandic cities, including Nuuk, the Capital, are supplied with electricity from hydropower plants.

Furthermore waste incineration plants have been established in Greenland. Some of the surplus heat from these plants is used for district heating, which to some extent reduces the need for fuel oil for heating and it reduces methane emissions as well.⁵

Greenland is eager to promote research into climate change⁵ and invites research projects to its territory in order to launch cooperation between Greenlandic and EU scientists.

The consequences of climate change appear first in the Polar Regions, and thus Greenlandic circumstances indicate what can be expected in the future on a worldwide basis. This makes Greenland an ideal place to carry out research and to serve as a living laboratory. It is hoped that the third International Polar Year 2007-2009 will help further the Greenlandic wish for research on the environmental effects of climate change, among other issues.

The International Polar Year is an event that takes place every 50 years. The purpose is to disseminate knowledge and appreciation of the fact that the Polar Regions interact with the oceans, atmosphere and landmasses, and that climate change in the Polar Regions indicates climate changes in the rest of the world.

THE NEED FOR FURTHER ACTION

The melting of the Greenlandic ice sheet influences the rising sea levels and may also affect the currents of the oceans. Both of these circumstances have consequences for European conditions and climate.

Immediate action is needed because climate change is not something that can be prevented. It is already

taking place and the local people are feeling the consequences in their everyday life.

As global society is to blame for climate change, it requires a global effort in order to combat the consequences. For these reasons Greenland encourages the EU to give scientists as well as civil servants the possibility of cooperating closely in matters related to the environmental effects of climate change in order to combat the most severe effects in the OCTs and their mother countries.



⁵) See Denmark's Climate Policy Objectives and Achievements, 2005

⁶) See speeches by Enoksen, Asii Narup Chemnitz and Alfred Jacobsen

CLIMATE CHANGE IN THE NETHERLANDS ANTILLES

The Netherlands Antilles consist of five islands in the Caribbean: Bonaire and Curaçao (the Leeward Group) in the south and Saba, St Eustatius and St Maarten (the Windward Group) in the Northeast.

All islands are surrounded by coral reefs. Higher seawater temperatures can induce “bleaching” in the corals – expulsion of the symbiotic algae, zooxanthellae. Bleaching causes a reduction in coral growth and over time causes the reefs to die. Coral reefs are important for coastal protection, biological diversity, fisheries and tourism. Dive tourism is important on all islands, but particularly on Bonaire. About 5 percent of the GDP of the Netherlands Antilles depends on coral reefs; for Bonaire and Saba it is much higher.

Sea level rise and ocean warming will change the general climate on the islands. Although irradiation from the sun in the tropics is high, temperatures are relatively mild because of cooling by wind and sea. The climate of the leeward group is already arid and it is still uncertain if it will get even more arid. But further desertification in Northern Africa may negatively influence the islands through more deposition of Sahara dust from Africa, which may include micro organisms, including pathogens, as well as reduced visibility and bronchial irritations.

THE HEATING OF THE OCEANS

Higher sea water temperatures will increase the probability of greater hurricane force and frequency. The island of the Windward Group in particular experience frequent hurricane with devastating results. If strong hurricanes become an annual event, the islands will not be able to cope anymore. The frequency of strong hurricanes in the Leeward Group is low presently, but that can change and swells and waves can destroy the marine habitats, which are important for tourism, as well as cause wider disasters.



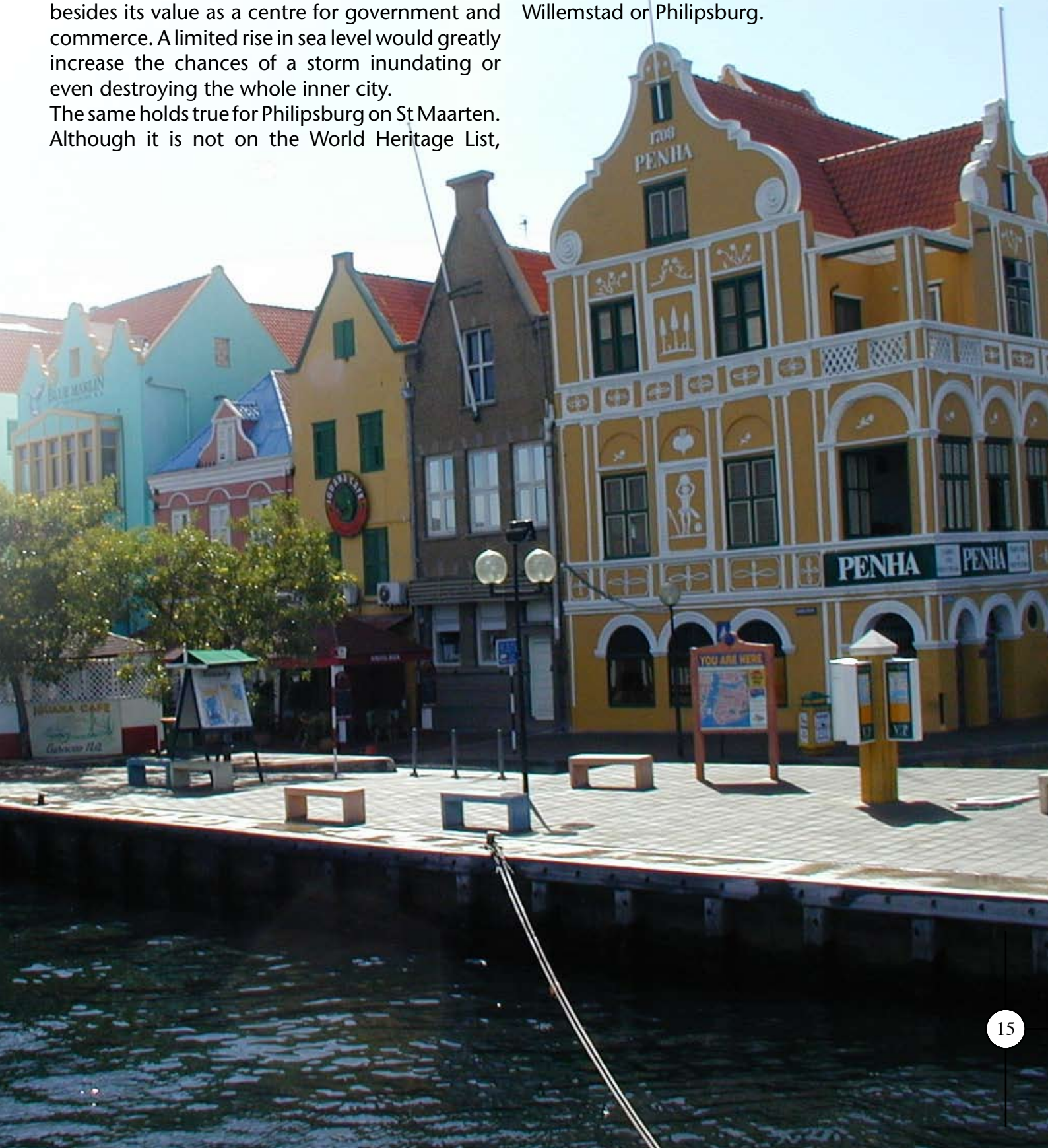
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THE RISING OF THE SEA LEVEL

The oldest part of Willemstad, the capital of the Netherlands Antilles on Curaçao, is built on a narrow low lying stretch. The city is declared a World Heritage Site and is one of the most important assets for tourism on the island, besides its value as a centre for government and commerce. A limited rise in sea level would greatly increase the chances of a storm inundating or even destroying the whole inner city.

The same holds true for Philipsburg on St Maarten. Although it is not on the World Heritage List,

the city lies in a low area and has a very active economy, particularly for cruise tourism. Other low lying towns in the Netherlands Antilles are Kralendijk on Bonaire and Simpson Bay on St Maarten. Damage by inundation in these towns could be considerable (many millions of euros), but in total economic loss still far less than in Willemstad or Philipsburg.





CLIMATE CHANGE IN ST HELENA



St Helena, an overseas territory of the United Kingdom (UK) over 4,000 miles away is a small island in the South Atlantic with an area of 47 square miles. Situated off the main shipping routes and at the moment without air access, the nearest land is Ascension Island, 700 miles distant. The nearest mainland is over 1,000 miles to the east in Angola, while Cape Town is 1,800 miles to the southeast and Brazil lies 1,800 miles to the west. The island is one of the most remote places in the world and as such is very vulnerable to outside influences.



CONSEQUENCES OF CLIMATE CHANGE

St Helena is home to a wide range of animal and plant species, many of which are found nowhere else in the world. These plants and animals are specially adapted to St Helena's unique climatic and other conditions. Whilst the extent to which their natural environments will be affected by the predicted changes in the climate is uncertain and more research is needed, it is highly likely that change in rain patterns and increased temperature will have a negative impact.



THE RISING OF THE SEA LEVEL AND THE HEATING OF THE OCEANS

The lack of commercially exploitable natural resources, a small domestic market and St Helena's physical location seriously limit potential for sustainable economic and industrial growth. The island is heavily dependant on UK Aid. It has a high import dependency and is totally reliant on transport by ships for carrying passengers, freight, and diesel and petrol fuel. The economy is weak, based mainly on offshore employment, agriculture and small fishing, tourism and building industries.

A 2001 report on the Impacts of Global Climate Change on the UK Overseas Territories, commissioned by DfID's Overseas Territories Unit and drawing on a study by the Natural Resources Institute, University of Greenwich and the Tyndall Centre for Climate Change Research, University of East Anglia, states : "Climate change impacts on fisheries interact with other stresses and are still poorly unders-



tood. Inshore fisheries are dependant on coastal ecosystems, dominated by reefs, mangroves and sea grasses. The largely negative impacts of sea level and sea temperature rises on environmental services that support fisheries represents a real and serious threat to near-shore fisheries. For deep-sea fisheries, the interaction between El Nino Southern Oscillation (ENSO) phenomena and fisheries production is a crucial area that is also currently poorly understood."

The current species of inshore fish could disappear due to not being able to adapt to temperature change and what little coral there is could also disappear for the same reason. Drastic decline in fish stock levels would seriously affect the export market with a negative impact on local fishermen's livelihoods and also the tourism attraction of the 'underwater world' around St Helena.

There is no breakwater or mooring facilities for ships, which have to anchor offshore. Containers' break bulk cargo and passengers are transferred between ship and shore on towed barges and small launches. Barges are held to the jetty for loading and offloading by ropes allowing for the rise and fall of the waves, which at times are too large for safe working. Rough sea conditions adversely impact on the economy when potential tourists fearing safety for life or limb are not able to venture ashore. Cruise-ships fearing risk to passengers have left because of this and some companies have taken St Helena off their itineraries altogether.

Sea level rise with more intense storms and stronger winds would exacerbate these conditions significantly, impacting negatively upon people's incomes and employment.

Failure to transfer fish destined for export would result in revenue loss from offshore. Presently there is no airport although plans are well in hand for St Helena to have an airport by 2010, funded by DfID.

INITIATIVES ALREADY TAKEN BY THE ST. HELENA GOVERNMENT

Currently St Helena depends almost entirely on diesel oil for generating electricity. Oil and also petrol fuel are required for motor vehicles, launches, electrical plant and equipment, and cooking. Fuel has to be imported and is carried to the island in fuel tankers. Although St Helena currently has three 80kW Lagerwey wind turbines, which between them can generate 20 percent of the island's electricity demand, unfortunately, the performance of these turbines has not been reliable. In January 2004 a report was received from a consultancy sponsored by DfID to review wind energy options for St Helena, and it is hoped that a DfID Renewable Energy Project will fund additional wind turbines.

The negative effects of climate change on the Forestry Sector are currently being addressed through the development of a rolling five-year management plan for the St Helena Government Forest Estate.

THE NEED FOR FURTHER ACTION

The predicted climate changes could mean additional stress on systems and services on which local livelihoods depend, impacting negatively upon employment, incomes, offshore revenue, budgets and health.

Heavier rain and more intense storms could have significant impact on island infrastructure including water systems, roads, power lines, housing and industrial plant.

Rock falls, which are already a problem when there is intense sun followed by heavy prolonged rains could increase, causing gullying of the hillsides, soil erosion, and increased damage to property and potentially lives. There could be additional risk from heat stress, and insect-borne and water-borne diseases may become prevalent as temperature and rainfall patterns change.

There is a need for action which St Helena can not by itself undertake. Cooperation with international partners, such as the EU, is thus welcome.



CLIMATE CHANGE IN ST PIERRE & MIQUELON

The Saint-Pierre and Miquelon archipelago lies in the North Atlantic Ocean, in the Gulf of Saint-Laurent, East of the Canadian coast and South of Newfoundland. It consists of several small islands, the two most important of which are Saint-Pierre, about 30 square kilometres, and Miquelon-Langlade, about 216 square kilometres, with 6,500 inhabitants.

The winter season starts towards November, with snow falling from December and sometimes lasting until May. In winter, the temperature mostly drops to between -3 and -10°C. It may fall to -15 to -20°C, mainly in the course of snowstorms with drifts, accompanied by strong winds.

The mild season begins towards June and goes on until October. However, June and July will often see quite a few days with fog. The average temperature during this season is about 18°C, sometimes reaching 20 or 25°C.

THE RISING OF THE SEA LEVEL AND THE HEATING OF THE OCEANS

Over the century in our archipelago, the Canadian Institute for Climate Studies foresees an average rise in temperature of about 3 degrees by 2080.

The main consequences of a change in the overall climate for Saint-Pierre and Miquelon will be:

- a possible increase in the number and intensity

of storms in winter;

- a probable intensification of cyclones in the Intertropical Zone, with cyclone trajectories veering eastwards, which the post-tropical depression phase could affect Saint-Pierre and Miquelon more often in summer and autumn;
- a warming of the ocean and alterations in acidity and salinity
- a rise in sea level.

A combination of these factors in our archipelago will bring about ever more frequent tidal storms and consequent flooding of the low-lying coastal areas, together with accelerated erosion of beaches and dunes on the coast. The Isthmus of Langlade and the Grand Barachois, which are wetlands with particularly rich but fragile ecosystems, could disappear as a result. A temperature increase of three degrees will certainly turn topsy-turvy the regional ecosystem, the archipelago's flora, fauna and the migratory movements of marine mammals and birds.

Bad weather will cause ever-worsening damage to dams and harbour infrastructures. More frequent and more intense tidal storms will force us to carry out large-scale operations to protect harbour areas and areas that could be flooded. This will pose a serious challenge to St Pierre and Miquelon's fragile economy. If the temperature, acidity and salinity of the ocean change, this will also undermine the marine ecosystem, and threaten the growth of fish stocks. What



will be the consequences for local fishing and aquaculture, currently in full swing? Will the sea product industry, which constitutes nearly all exports, be able to survive a new deterioration of exploitable resources?

INITIATIVES ALREADY TAKEN BY THE ST PIERRE AND MIQUELON GOVERNMENT

In addition to regular awareness-raising activities to influence public opinion about sustainable development and global warming, the local authorities have developed windmill power-

generating facilities, making it possible to reduce the production of diesel-based electricity and to limit the volume of greenhouse gases.

THE NEED FOR FURTHER ACTION

Today, in order to limit the scope of climate warming and its consequences, rapid action to reduce greenhouse gas emissions is necessary both at local level and by international organisations.

CLIMATE CHANGE IN THE TURKS AND CAICOS ISLANDS

The Turks and Caicos Islands (TCI) are made up of a series of low lying limestone islands that are an extension of the south eastern Bahamas Platform. The Islands rest on two shallow limestone banks separated by a 35.4 km wide deep-water channel, the Turks Island Passage. Forty islands and cays make up the TCI, however only eight are inhabited : Grand Turk, Salt Cay, South Caicos, Middle Caicos, North Caicos, Parrot Cay, Pine Cay and Providenciales. The total landmass of the islands is 193 square miles. The estimated population is 32,000.

The islands in TCI are small and more than half the islands' land mass is either below sea level or just a few metres above sea level. As such more than 70 % of the settlements are located on coastal lowland.

The people of these islands depend heavily on limited natural resources for their very existence. Tourism, based mainly on the sun, sand and sea is the mainstay of the economy and a sizable number of people also rely on fisheries as a source of food and income.

THE HEATING OF THE OCEANS

The Turks and Caicos Islands are subjected to the impacts of global warming. Most marine organisms live within narrow temperature regimes, and even short-term extreme temperature increase can have a dramatic impact. Over the past two decades, for example, short-term extreme high temperatures have contributed to a decline of coral reefs throughout the region. Corals, stressed by high temperatures, may eject their symbiotic algae and become bleached. This phenomenon renders the corals less able to cope with additional physiological stress and many of the colonies can die. 2005 was the warmest year ever recorded, beating the previous record high set in 1998. As a consequence, reefs across the islands were severely bleached, and as many as 75 % of the coral areas were impacted. Luckily for the TCI the majority of corals affected have regained normal coloration and no major die-

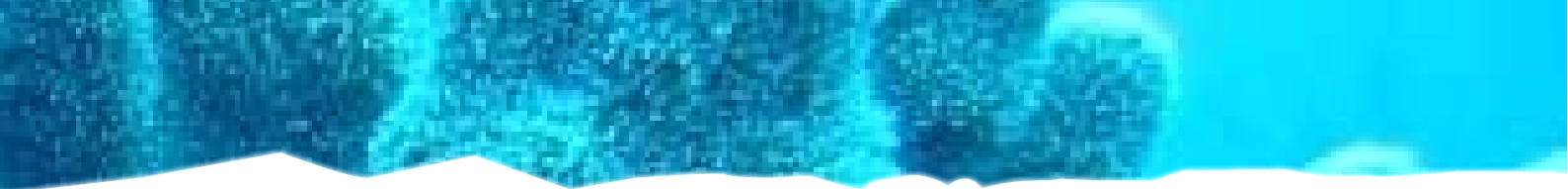
offs have been reported.

Coral bleaching can affect coastal fisheries. When the temperature increases by one or two degrees, the coral begins to die and fish populations decrease. This harms not only fishers' livelihoods, but also the tourism industries derived from the coral reefs.

THE RISING OF THE SEA LEVEL

In the TCI, the people, tourist resorts and infrastructure (including roads and airports) are concentrated in the coastal zones, and are thus especially vulnerable to any rise in sea level.

Global warming is predicted to cause a 1-2 cm per year rise in sea level during this century. This is a direct threat to the TCI as all low lying coastal areas would disappear, including the beaches, hotel areas and a majority of the settlements.



Storm surges associated with increased hurricane events will bombard coastal areas and wash away beaches and coastal properties.

INITIATIVES ALREADY TAKEN BY THE TCI GOVERNMENT

Planning Regulation was updated in 1998 to include a 100 ft setback for coastal development. A coastal vulnerability study is being undertaken.

Shaded relief maps of the Islands highlighting areas vulnerable to flooding were recently developed. This will aid in the overall disaster management planning.

A Sand Study has been initiated to identify potential sources of beach quality sand to replenish beach areas in the event of beach loss due to hurricanes and / or sea level rise.

Reef restoration strategy is on the way. This involves a rapid assessment of the health of the coral reefs in the TCI to identify reef areas needing restoration. The strategy will also determine the appropriate reef restoration technologies, particularly exploring the BioRock™ technology which has been reported to facilitate faster growth rates in corals and increase survivorship to temperatures up to fifty times higher than the natural reefs.

THE NEED FOR FURTHER ACTION

Failure to adapt to global warming and climate change could lead to grave socio-economic costs. The economic disruption could be devastating, even to the extent of requiring population relocation into other islands or forcing people to emigrate from the islands.





CONCLUSION

From the Tropics and the Polar OCTs

Climate change is one of the greatest environmental, social and economic threats facing the planet. We are all responsible for climate change but are affected differently.

The vulnerability of islands and littorals was recognized by the United Nation General Assembly in 1989. 17 years later no specific measures have been implemented for OCTs, and OCTs as a whole still remain too isolated from international dialogues and policies. The European Union is at the forefront of international efforts to combat climate change. However OCTs require a stronger voice on the European and international scene. A global effort is needed to combat climate change and its impacts. The oceans are a good illustration of why climate change is a concern for the whole world and requires better cooperation between the OCTs and the European Union.

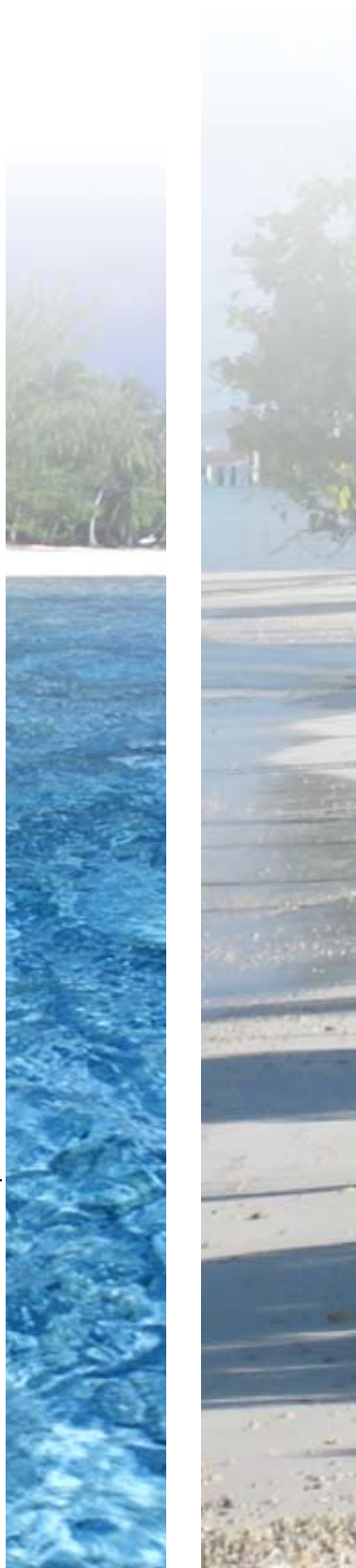
SUGGESTED ACTIONS

Research is a vital tool!

Research into climate change in the OCTs is very useful. It has been claimed that climate change in the Arctic is felt 2-3 times more than in the rest of the world and that the island populations in all low lying territories face the magnified risks of sea-level rise, hurricanes, floods and tsunamis. Developing mechanisms that enable the transfer of available and acquired knowledge will help to formulate answers to OCTs' needs. Particular attention needs to be paid to how scientific knowledge can be accessed and then integrated into decision-making processes.

While the OCTs have limited research resources themselves, they represent considerable potential for global research. It is thus it is of great importance that OCTs improve their use of, and access to, EU's Framework Programme for Research and Development.

OCTs would recommend that the EU allows OCTs access to the EU's environmental programmes such as LIFE+, which among other things would enable the OCTs to cooperate with European researchers in projects regarding issues of common concern.





to the European Union

Participation in international conferences

The OCTs want to contribute to the international debate on climate change, but it is difficult for small actors to get a political message through to the world arena. It would be helpful to allow the OCTs to be represented at international conferences such as within the UN. Exchange of knowledge and ideas in an international forum could create a foundation for global cooperation on combating climate change.

Strategy towards sustainable development with an environmental focus

EU programme access is one tool. However existing cooperation could provide further mechanisms. The OCTs, the Commission and the four Member States already meet in Partnership Working Parties and could build on the work done by adopting a strategy towards a better environment by 2011. The OCTs urge the trilateral meetings to work towards an environmental action plan.

The OCTs recommend setting out an agenda for dealing with climate change in the OCTs by developing an EU-OCT vision for the oceans and seas. As the EU points out in its Maritime Policy paper : "Oceans and seas cannot be managed without cooperation with third countries and in multilateral fora. EU policy aimed at the oceans must be developed within that international context".





OCTAssociation

This brochure was produced by OCTA (Association of Overseas Countries and Territories of European Union) with contributions by OCTA members for the 5th OCT-EU Forum in NUUK, Greenland.
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